MoTrans-15: 14-Day Implementation Plan

Monetary transmission to equities across the 15 largest economies: policy surprises, central bank language and blind transfer to an unseen cycle

1. Purpose and honest scope

This plan compresses the project into a 14-day sprint using AI-assisted coding (Claude Code) plus a small human team. The realistic output is a complete first-pass dataset, core results, the blind-transfer test and a full manuscript draft.

Not realistic in 14 days: paid intraday data, full speech-level language coverage, and a verified, submission-ready paper. The careful version of this project takes about 8 months. The sprint trades breadth for a working core, and every cut is listed below so nothing is hidden.

Core claims (in scope)

1. Blind transfer: a transmission relationship estimated on pre-2020 data predicts the 2022-2025 tightening cycle and held-out countries.
2. Incremental language: central bank tone adds information beyond the rate surprise.
3. State dependence: valuations, bank exposure and FX regime explain which markets respond more.

Out of scope (future work): spillover network or graph model, counterfactual simulator, portfolio demo, intraday data. Hierarchical Bayes is a stretch goal only. Fed-to-others spillover is limited to one regression.

2. Locked design decisions (freeze on Day 1)

Item	Decision
Countries	US, China, Germany, India, Japan, UK, France, Italy, Russia, Canada, Brazil, Spain, Mexico, Australia, South Korea (top 15 by IMF 2026 nominal GDP)
Policy regimes	12 independent central banks: Fed, ECB (serves DE, FR, IT, ES), PBoC, BoJ, BoE, CBR, BoC, BCB, Banxico, RBA, BoK, RBI. Euro area is ONE shock source with four market responses

Item	Decision
Data window	2000 (or earliest reliable date, about 2005 for some emerging markets) to latest available. 2011 onward is a robustness subsample only, because 2011-2019 has too few tightening events
Russia	Flagged after Feb 2022 (sanctions, trading halts, capital controls). Every analysis must run with and without it
Indices	3 per country: broad, banks/financials, small-cap or growth (table in the concept note). Euro-area bank index is not country-specific, so verify and document
Frequency	Daily. Response windows: 1, 3, 5, 20 and 60 trading days
Returns	Log return of the local-currency price index (primary); USD return as robustness
Treatment	Standardized policy surprise (unit: one standard deviation within each shock source). Decisions include hikes, cuts and holds
Shocks	US and euro area: public surprise series. Other regimes: announcement-day change in the 1-year government yield (proxy). Each country gets a shock-quality grade A/B/C
Timing rule	Each decision is mapped to its announcement time and time zone, then to the FIRST trading session of each exchange that can react (for example a Fed decision reaches Japan, India and Korea the next local day)
Split	Train: up to Dec 2019. Embargo/crisis: 2020-2021 reported separately. Blind test: 2022 to latest. Plus leave-one-country-out

3. Roles

Role	Responsibility
Project lead	Owns the spec, approves each gate, writes the paper in the team's own voice
Data lead	Sources data, checks licences, spot-checks values against original sources
Text lead	Collects statements, hand-labels the validation sample
Modeling lead	Reviews model code, runs final analyses, checks for leakage
Supervisor	Reviews the pre-registration (Day 1) and the first results (Day 9)
Claude Code	Writes and debugs code, tests, documentation, drafts text. Does NOT download unverified data silently, invent references or decide claims

4. Timeline at a glance

Day	Phase	Main output
1	0. Setup and pre-registration	Repo, environment, locked spec, source registry
2-3	1. Pilot (US, Germany/euro area, India, Japan)	Working pipeline and timing tests
4-6	2. Full dataset (15 countries, about 45 indices)	Harmonized panel and quality report
5-7	3. Shock construction	Shock dataset with quality grades
5-8	4. Language layer	Meeting-level tone features
8-9	5. Baseline econometrics	Local projection results
10-11	6. Causal ML	DML and causal forest results
12	7. Blind validation	Headline generalization result

Day	Phase	Main output
13	8. Robustness and light spillover test	Placebos, alternatives, Fed regression
14	9. Release and final assembly	Manuscript draft, repo, data dictionary

Manuscript writing runs in parallel from Day 8 (data and methods sections can be written as soon as Phases 2-4 pass their gates).

5. Phase details

Phase 0: Setup and pre-registration (Day 1)

Tasks

- Create repo: data/raw (read-only), data/interim, data/final, src, tests, notebooks, docs, outputs; pinned environment file; task runner so the whole pipeline runs with one command.
- Write docs/preregistration.md: hypotheses, shock definitions, outcome definitions, windows, split design, placebo tests, and which tests are primary versus exploratory.
- Write docs/source_registry.md (every variable, source, access method, known gaps, licence) and docs/data_dictionary.md.
- Start DECISIONS.md, a dated log of every choice.

Gate: supervisor and team approve the pre-registration. Freeze it. Later changes are logged as deviations.

Phase 1: Pilot, 4 markets (Days 2-3)

Tasks

- Collect policy rates and decision dates for Fed, ECB, RBI, BoJ; 3 indices each; daily returns.
- Build the announcement-time alignment module and event-window extractor.
- Unit-test on at least 5 known events per market (for example the March 2022 Fed hike, March 2020 emergency cuts).

Gate: hand-check 10 events against original sources; timing rule behaves correctly for every time zone. Do not scale up until this passes.

Phase 2: Full dataset (Days 4-6)

Tasks

- Extend to all 15 countries and about 45 indices, in batches, running data checks after each batch.
- Add controls: 1-year yields, FX, volatility measures, CPI, growth indicators, valuation proxies (P/E or earnings yield where available).
- Record every index start date, gap, outlier and substitution. If an index has no usable free source, log it and replace it with the closest alternative; never fill silently.

Output: harmonized daily panel plus a data-quality report (missing dates, outliers, coverage by country). **Gate**: no unexplained gaps; provenance and licence recorded for every series; Data lead spot-checks at least 20 values per country group.

Phase 3: Shock construction (Days 5-7)

Tasks

- US and euro area: load public surprise series and extend them to the latest meetings where possible.
- Other regimes: build the yield-change proxy on decision days, standardize within each source, assign grade A (clean public series), B (reasonable proxy) or C (weak).
- Validate against known surprise episodes (for example the 2022 Fed hikes; 2013 taper tantrum period).

Gate: shock series behave sensibly on known episodes. Grade C countries are descriptive-only in the causal analysis.

Phase 4: Language layer (Days 5-8)

Scope for the sprint: official policy statements for central banks with English statements (Fed, ECB, BoE, BoJ, RBI and others where available), plus BIS speeches where easy to align. Document coverage honestly; the language claim applies to this subset.

Tasks

- Collect and align each statement to its decision date.
- Score tone with FinBERT and a structured LLM classification (hawkish/dovish, inflation concern, growth concern, uncertainty, forward-guidance shift).
- Text lead hand-labels about 150-200 statements; compare to model scores and report agreement.

Gate: model scores agree reasonably with hand labels and with known hawkish/dovish meetings. If agreement is poor, use the LLM-only score and report it.

Phase 5: Baseline econometrics (Days 8-9)

Model: panel local projections, for each horizon h: cumulative return from t to t+h = index fixed effect + beta_h x shock_t + controls (lagged state variables) + error.

- Standard errors clustered by event date, because euro-area markets share one shock and are not independent observations.
- Estimate by index class (broad, banks, small-cap), by advanced vs emerging economies, with and without Russia.

Gate: results stable across windows; supervisor reviews the first results on Day 9.

Phase 6: Causal ML (Days 10-11)

- Double ML with time-ordered cross-fitting (never random folds); causal forest for heterogeneous effects; quantile models for left-tail response; conformal intervals for calibrated uncertainty; SHAP for interpretation.
- Language test: compare shock-only model with shock plus language model, out of sample.
- Stretch only: hierarchical Bayesian pooling across countries.

Gate: ML estimates are consistent with or clearly improve on the baseline; any gain is out-of-sample, not in-sample.

Phase 7: Blind validation (Day 12)

Run exactly as pre-registered:

- Train up to Dec 2019, predict the 2022 to latest cycle blind.
- Leave-one-country-out (train on 14 economies, test on the 15th; treat euro-area markets as one block so the test is not contaminated).
- 2020-2021 reported separately as a crisis regime.
- Metrics: out-of-sample sign accuracy, out-of-sample R-squared against a no-shock benchmark, conformal interval coverage, stability of beta_h between training and test periods.

Rule: report failures. A well-analysed failure that shows where transfer breaks is a legitimate result.

Phase 8: Robustness and light spillover test (Day 13)

- Placebo tests: artificial dates and non-policy days.
- Alternative shocks (yield proxy vs public surprise), alternative windows, USD returns, 2011+ subsample, with and without Russia.

- Multiple-testing correction on the heterogeneity results.
- One spillover regression: Fed shock to the other markets, split advanced vs emerging.
- Ask Claude to audit for leakage, timestamp errors and alternative explanations; the Modeling lead verifies every finding.

Phase 9: Release and final assembly (Day 14)

- Manuscript draft following the blueprint in the concept note (abstract, introduction, related work, data, identification, results, mechanisms, generalization, robustness, limitations).
- Repo with code, data dictionary, provenance log and instructions to reproduce.
- Release only derived variables and code where licences forbid raw price redistribution.
- Run the full pipeline on a clean machine as a reproducibility test.

6. Working rules for Claude Code

- Work phase by phase. At the end of each phase, stop, summarize, show test results and open problems, and wait for human approval.
- Never fabricate or silently interpolate data. If a source fails, log it and ask.
- Every dataset carries provenance: source, URL or API, retrieval date, licence, transformations.
- No look-ahead leakage; no random shuffling of time series.
- Do not invent citations. The team builds the bibliography only from sources it has opened.
- Do not state causal claims beyond what a proxy shock supports.

7. Risks and fallbacks

Risk	Fallback
Index series unavailable on free sources	Use closest alternative, document, or drop from the causal analysis
Weak shock proxy for a country	Grade C, descriptive only
Few hike events per country	Pool across countries, use all policy changes, cluster by event
Language coverage limited to English statements	Restrict the language claim to that subset and say so

Risk	Fallback
Blind test fails	Report where and why transfer breaks; do not tune on the test period
Behind schedule	Cut stretch items first (hierarchical Bayes, spillover regression), never cut validation or provenance
Claude usage limits hit	Prioritize Phases 1-3 and 7; check plan limits early

8. Governance and ethics

- Pre-register and freeze the specification; log all deviations.
- Disclose AI-tool use in the methods section per Springer Nature policy; AI tools cannot be authors. Check the target journal's current policy.
- Respect data licences; do not publish proprietary price series.
- Humans verify every number in the paper against outputs, and every reference against the original source.

9. Definition of done

- [ ] Harmonized dataset with provenance, data dictionary and quality report
- [ ] Shock series with quality grades
- [ ] Language features with validation against hand labels
- [ ] Baseline, causal ML and blind-transfer results with uncertainty
- [ ] Placebo and robustness suite
- [ ] Reproducible repository passing a clean-machine run
- [ ] Full manuscript draft with every number and citation verified

After the sprint: expect the draft to need several more weeks of verification, supervisor feedback and revision before submission to Nature Communications or Scientific Data.